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Order–disorder transition of β-brass (CuZn) by Metropolis Monte Carlo on a rigid lattice

Tony Pansera Noé Fourniol

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Abstract

We wrote a Metropolis Monte Carlo code that exchanges Cu and Zn atoms on a fixed body-centred cubic lattice of 432 sites, with a pair energy model limited to first and second neighbours. The code reproduces the exact values of the perfect B2 structure (cohesive energy 2.484 eV/atom) and of a random alloy. Scanning the temperature from 200 K to 1600 K, we follow the energy, the heat capacity, the long-range order parameter η, its susceptibility and the Cowley short-range order parameters. The three estimates of the transition temperature all give $T_c \approx 720$ K, close to the experimental value of about 740 K and well below the Bragg–Williams mean-field value of 894 K. Short-range order survives above T_c.

1 Introduction

β-brass is the CuZn alloy with about 50 % Zn. At low temperature it is ordered in the B2 (CsCl-type) structure: the atoms sit on a body-centred cubic (bcc) lattice, Cu on the cube corners and Zn on the cube centres. When the temperature rises, Cu and Zn atoms swap places more and more often. Around 740 K the distinction between the two kinds of sites disappears: the alloy becomes a disordered bcc solid solution [1–3]. This is a classic order–disorder transition, first described by Bragg and Williams with a mean-field theory [4].

In this project [5] the atoms do not move away from the lattice sites ("rigid lattice"): only the occupation of each site by Cu or Zn changes. The energy is a sum of pair interactions between first and second neighbours. We sample the equilibrium configurations with the Metropolis algorithm [6] and measure how the order is lost when the temperature increases.

The report follows the questions of the project. Section 2 describes the model and the algorithm, and Section 3 gives all the simulation parameters. Section 4 tests the program and gives the cohesive energy. Section 5 shows how long the system takes to reach equilibrium. Section 6 presents the physical quantities as a function of temperature, Section 7 the transition temperature and Section 8 the discussion.

2 Model and method

2.1 Lattice and neighbours

The simulation box contains $6 \times 6 \times 6$ cubic cells of side a, with two sites per cell (Figure 1): a site of sublattice α at the corner $(0, 0, 0)$ and a site of sublattice β at the centre $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. This gives $N = 2 \times 6^3 = 432$ sites, with periodic boundary conditions in the three directions. Each site has 8 first neighbours at distance $\sqrt{3}a/2$, all on the other sublattice, and 6 second neighbours at distance

a, on the same sublattice. The two neighbour tables are built once at the start of the program: the periodic boundary conditions are applied by taking each cell coordinate modulo 6.

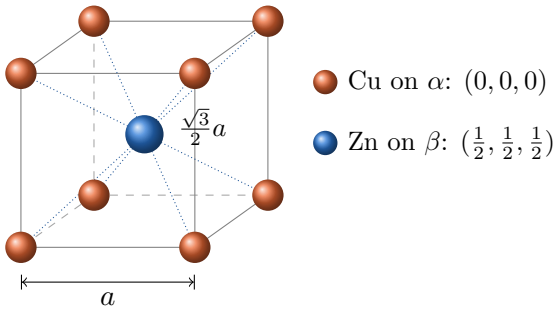


Figure 1: Cell of the B2 structure of CuZn (perfect order, $T = 0\,\mathrm{K}$). Cu atoms (orange) occupy the corners (sublattice α), Zn atoms (blue) the centre (sublattice β). The dotted lines are the 8 first-neighbour bonds of the centre atom (length $\sqrt{3}a/2$); second neighbours are along the cube edges (length a). In the disordered phase, Cu and Zn occupy both sublattices at random, and the lattice is simply bcc.

2.2 Energy

The energy of a configuration is

$$E = \sum_{\langle i,j\rangle_1} J_{\sigma_i\sigma_j} + \sum_{\langle i,j\rangle_2} I_{\sigma_i\sigma_j}, \tag{1}$$

where $\sigma_i \in \{\mathrm{Cu}, \mathrm{Zn}\}$ is the atom on site i, and the sums run over first- and second-neighbour pairs, each pair counted once. The parameters are given in Table 1. In the program, the energy of each site with its 14 neighbours is summed over all sites and divided by 2, since every pair is seen twice.

Table 1: Pair energies of the model [5], in eV. J acts between first neighbours, I between second neighbours. The last column is the "ordering energy" $w = X_{\mathrm{CuCu}} + X_{\mathrm{ZnZn}} - 2X_{\mathrm{CuZn}}$ of each shell: $w > 0$ favours unlike pairs, $w < 0$ favours like pairs.

Shell	Cu–Cu	Zn–Zn	Cu–Zn	w
1st neighbours (J)	-0.873	-0.338	-0.621	$+0.031$
2nd neighbours (I)	0.000	0.000	$+0.005$	-0.010

The first shell favours Cu–Zn pairs ($w_1 > 0$) and the second shell favours pairs of the same species ($w_2 < 0$). In the B2 structure the 8 first neighbours of every atom are of the other species and the 6 second neighbours of the same species, so both shells push towards B2 order.

2.3 Monte Carlo moves

The composition is fixed (216 Cu, 216 Zn), so a move must exchange two atoms rather than change the species of one site (Kawasaki dynamics [7]). One Monte Carlo step is:

1. draw a site i at random, then one of its 8 first neighbours j at random;
2. if i and j hold the same species, nothing changes; otherwise compute the energy change ΔE of the exchange;
3. accept the exchange with the Metropolis probability $P = \min\left(1, e^{-\Delta E/k_B T}\right)$ [6].

ΔE only involves the bonds of i and j with their other neighbours (7 first and 6 second neighbours each); the i–j bond itself stays a Cu–Zn bond. This local update costs 26 terms instead of the

$432 \times 14/2 = 3024$ bonds of the full sum. The energy is then updated incrementally after every accepted move. A *sweep* is $N = 432$ attempted exchanges, so that every atom is chosen once on average.

2.4 Measured quantities

After equilibration, we record once per sweep the energy E, the long-range order parameter and the two Cowley parameters:

$$\eta = \frac{N_{\mathrm{Cu}}^{(\alpha)} - N_{\mathrm{Cu}}^{(\beta)}}{N_{\mathrm{Cu}}^{(\alpha)} + N_{\mathrm{Cu}}^{(\beta)}}, \qquad \alpha_n = 1 - \frac{p_n^{\mathrm{Zn|Cu}}}{c_{\mathrm{Zn}}} \quad (n = 1, 2), \tag{2}$$

where $N_{\mathrm{Cu}}^{(\alpha)}$ is the number of Cu atoms on sublattice α, $p_n^{\mathrm{Zn|Cu}}$ the probability that a neighbour of shell n of a Cu atom is a Zn atom, and $c_{\mathrm{Zn}} = 1/2$ [8]. The perfect B2 order gives $\eta = 1$, $\alpha_1 = -1$ and $\alpha_2 = +1$; a random alloy gives $\eta = \alpha_1 = \alpha_2 = 0$. In a finite box the ordered state can also form with Cu on β ($\eta < 0$), and near T_{c} the system goes back and forth between the two. We therefore average $|\eta|$, which measures the amount of order whatever its sign.

From the series of M measurements we compute the means $\langle E \rangle$, $\langle |\eta| \rangle$, $\langle \alpha_n \rangle$, and two response functions from the fluctuations [9]:

$$\frac{C}{N k_B} = \frac{\langle E^2 \rangle - \langle E \rangle^2}{N\,(k_B T)^2}, \qquad \chi = \frac{N\,(\langle \eta^2 \rangle - \langle |\eta| \rangle^2)}{k_B T}. \tag{3}$$

The heat capacity C and the susceptibility χ (in eV^{-1}) both have a peak at the transition, where the fluctuations are largest.

3 Simulation conditions

Table 2 lists everything needed to redo our calculations. Each temperature is an independent run that starts from the perfect B2 structure.

Table 2: Conditions of all the calculations of this report. The numbers of steps refer to attempted exchanges; 1 sweep = 432 steps. The full run (equilibration test and scan) is done by one script, `report/make_figures.py` (Section B).

Lattice	bcc, $6 \times 6 \times 6$ cubic cells, $N = 432$ sites, periodic boundary conditions
Composition	216 Cu + 216 Zn ($c_{\mathrm{Zn}} = 1/2$), conserved by the moves
Energy	pair model of Equation (1), parameters of Table 1; $k_B = 8.617\,333\,262 \times 10^{-5}$ eV/K
Moves	exchange of a random site with a random first neighbour, Metropolis acceptance
Starting configuration	perfect B2 (Cu on α, Zn on β) at every temperature
Equilibration test	200 000 steps (463 sweeps) from B2 at each temperature of the scan, energy recorded after every step
Equilibration	$N_{\mathrm{eq}} = 50\,000$ steps (116 sweeps), not measured
Measurement	$M = 4000$ sweeps, one measurement of E, η, α_1, α_2 per sweep
Temperatures (K)	200, 300, 400, 500, 550, 600, 650, 675, 700, 710, 720, 730, 740, 750, 760, 775, 800, 850, 900, 1000, 1200, 1600
Random numbers	Python `random` module (Mersenne Twister [10]); seeds 2026 (checks), 2027 (equilibration test), 2028 (scan); NumPy seed 2026 for the random lattices
Software	Python 3.12.13, NumPy 2.5.4, matplotlib 3.11.2
Hardware and cost	one core of an AMD Ryzen 7 6800U laptop; 6.7 µs per step, 3.5 ms per measured sweep; 366 s for the equilibration test and the scan

4 Tests of the program and cohesive energy

Table 3 gathers the tests of the program. The neighbour tables are correct, the perfect B2 structure gives the exact values, and random lattices give no order. For a random lattice $\langle|\eta|\rangle$ is not zero but about 0.04: with only 216 Cu atoms, chance alone puts a few more of them on one sublattice. The number of Cu on α follows a hypergeometric law, which gives the value $\sqrt{2/\pi}\,\sigma_\eta = 0.038$; this is the lowest value $\langle|\eta|\rangle$ can reach in our box. Finally the energy updated move by move stays equal to the energy recomputed from scratch, which validates the ΔE formula.

Table 3: Tests of the program. Random lattices are obtained by shuffling the 432 atoms (2000 lattices for η, 200 for the Cowley parameters). The energy test is a run of 10^5 steps at 730 K, after which the energy tracked step by step is compared with a full recomputation of Equation (1).

Test	Expected	Obtained
Neighbour tables	8 and 6 distinct neighbours, symmetric; 1st on the other sublattice, 2nd on the same	all true
B2: energy per atom	$4J_{\mathrm{CuZn}} = -2.484\,\mathrm{eV}$	$-2.4840\,\mathrm{eV}$ ($E = -1073.088\,\mathrm{eV}$)
B2: η, α_1, α_2	1, −1, +1	1.000, −1.000, +1.000
Random lattices: α_1, α_2	0, 0	−0.002, −0.004
Random lattices: $\langle\|\eta\|\rangle$	$\sqrt{2/\pi}\,\sigma_\eta = 0.038$	0.038
Tracked vs recomputed energy	0	$4 \times 10^{-12}\,\mathrm{eV}$
Number of Zn after the run	216	216

Cohesive energy. At $T = 0\,\mathrm{K}$ the system is in the B2 structure. Each atom has 8 first-neighbour Cu–Zn bonds, each shared by two atoms, and its second-neighbour bonds cost nothing ($I_{\mathrm{CuCu}} = I_{\mathrm{ZnZn}} = 0$). The energy per atom is therefore $E/N = 4J_{\mathrm{CuZn}} = -2.484\,\mathrm{eV}$, which is what the program finds. Since the zero of energy corresponds to separated atoms, the cohesive energy of B2 CuZn is $E_{\mathrm{coh}} = 2.484\,\mathrm{eV/atom}$. With the same model, pure Cu and pure Zn on a bcc lattice would have $4|J_{\mathrm{CuCu}}| = 3.492\,\mathrm{eV/atom}$ and $4|J_{\mathrm{ZnZn}}| = 1.352\,\mathrm{eV/atom}$, which are the experimental cohesive energies of copper (3.49) and zinc (1.35 eV/atom) [11]. The ordered alloy is therefore $0.062\,\mathrm{eV/atom}$ more stable than the mean of the two pure metals.

5 Equilibration

Every run starts from the perfect B2 structure, which is not representative of a finite temperature. The first steps must be discarded until the system has "forgotten" this initial state. To choose this number of steps, we record the energy after every step during 200 000 steps at each temperature of the scan (Figure 2, left). The equilibrium energy E_{eq} and the size σ of its fluctuations are the mean and standard deviation over the second half of the run; equilibrium is reached at the first step where $E \geq E_{\mathrm{eq}} - \sigma$.

At low temperature (200–300 K) the B2 structure is already the equilibrium state and almost no exchange is accepted. At high temperature, disorder sets in within a few thousand steps. The slowest relaxations are found below and near the transition, with at most 11 806 steps (27 sweeps) at 730 K. This estimate is itself noisy, because the energy fluctuates strongly near T_{c}. We therefore discard $N_{\mathrm{eq}} = 50\,000$ steps at every temperature, more than four times the slowest relaxation.

6 Physical quantities versus temperature

Figure 3 shows what the configurations look like. At 300 K the layer is a perfect checkerboard of Cu and Zn. Near T_{c} the order is only partial, and at 1200 K the atoms are spread at random over the

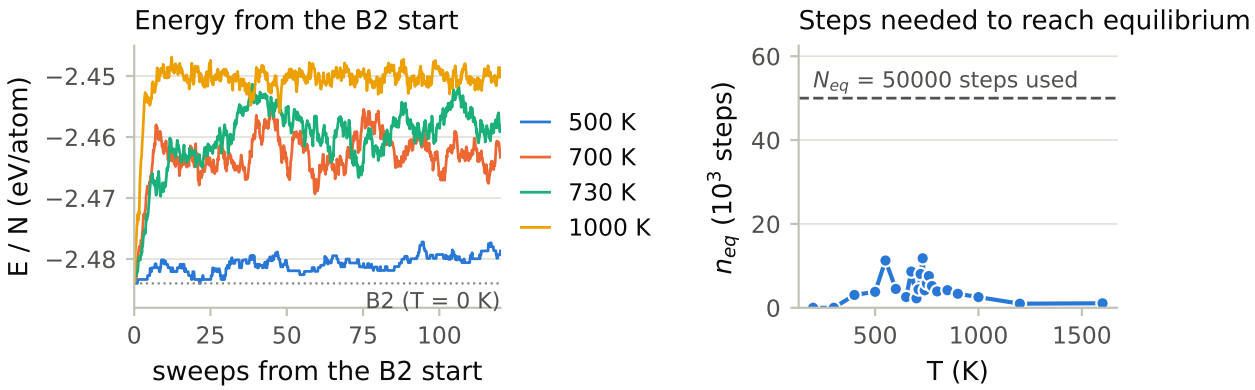


Figure 2: Equilibration from the perfect B2 structure. Left: energy per atom during the first 120 sweeps at four temperatures (one value every 50 steps); the dotted line is the B2 energy. Right: number of steps n_{eq} needed to reach equilibrium at each temperature of the scan (first step with $E \geq E_{\mathrm{eq}} - \sigma$, see text); the dashed line is the number of steps $N_{\mathrm{eq}} = 50\,000$ discarded in all the runs.

two sublattices.

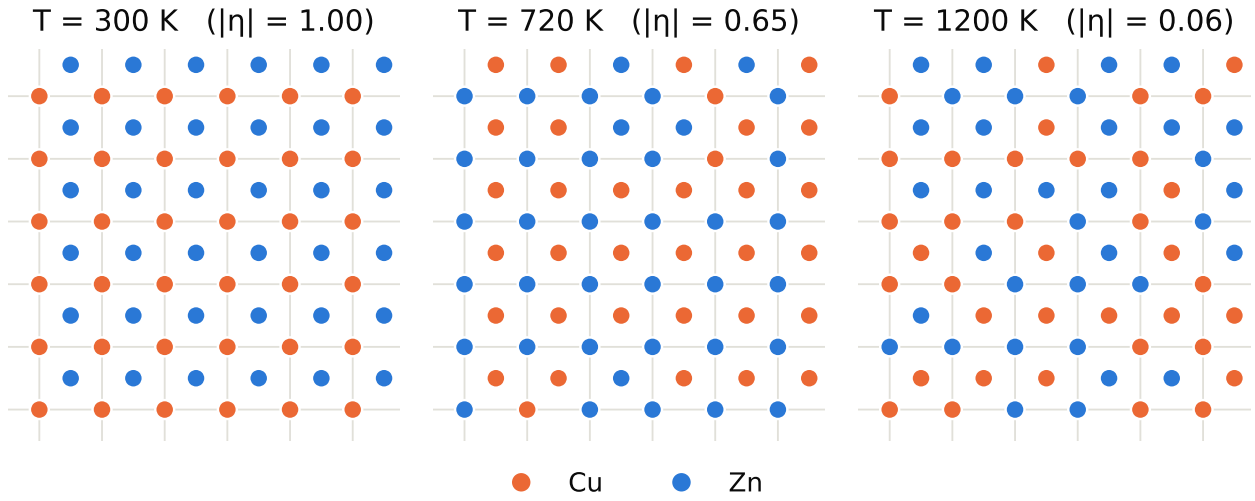


Figure 3: Final configuration of the runs at three temperatures, seen from above: one layer of the box, with the α sites of height $z = 0$ at the grid nodes and the β sites of height $z = a/2$ at the centres of the squares (6×6 cells, 72 of the 432 atoms). $|\eta|$ is computed on the whole box.

Energy and heat capacity. The mean energy (Figure 4, left) starts from the B2 value and rises steeply between 600 and 800 K, where most of the disorder appears. At high temperature it tends towards the energy of the ideal random alloy, $E_{\mathrm{dis}}/N = 4\bar{J} + 3\bar{I} = -2.4455\,\mathrm{eV/atom}$, where $\bar{J}$ and $\bar{I}$ are the averages of the pair energies weighted by the probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$ of Cu–Cu, Zn–Zn and Cu–Zn pairs. Even at 1600 K the energy remains 0.0023 eV/atom below this value, because of the short-range order discussed below. The heat capacity (Figure 4, right) has a clear peak at 720 K; its value at 700 K is almost the same, so the top of the peak is between 700 and 720 K.

Long-range order. $\langle|\eta|\rangle$ (Figure 5, left) decreases slowly up to about 600 K, then drops fast between 650 and 800 K, and finally reaches the finite-size floor of the random lattice. In an infinite crystal η would vanish exactly at T_{c}; in our small box the drop is rounded. The susceptibility (Figure 5, right) has a sharp peak at 720 K, almost equal to its value at 730 K.

Figure 6 explains these peaks. Below T_{c} the distribution of $|\eta|$ is narrow around a large value; above T_{c} it is narrow close to zero. Near T_{c} it is very broad: the system wanders between ordered and

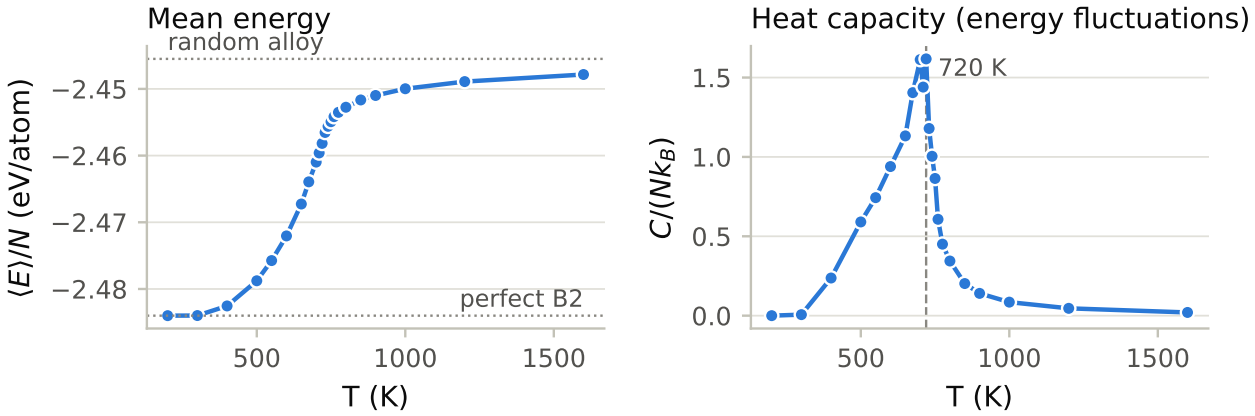


Figure 4: Left: mean energy per atom versus temperature; the dotted lines are the perfect B2 energy (-2.484 eV/atom) and the energy of an ideal random alloy (-2.4455 eV/atom). Right: heat capacity per atom in units of k_B, from the energy fluctuations (Equation (3)); the dashed line marks its maximum.

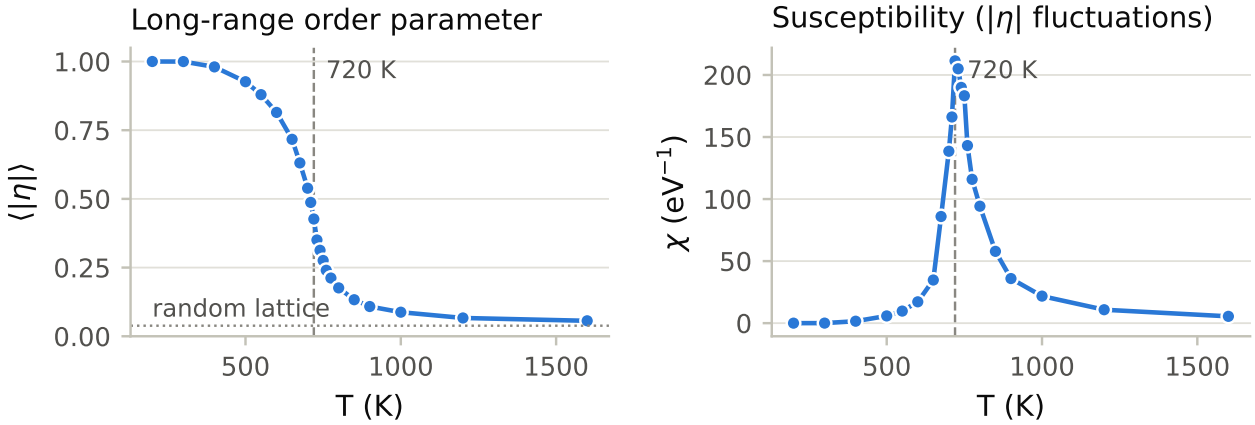


Figure 5: Left: mean absolute long-range order parameter; the dotted line is the value for random lattices (finite-size floor), the dashed line marks the steepest drop. Right: susceptibility of the order parameter from the fluctuations of $|\eta|$ (Equation (3)); the dashed line marks its maximum.

disordered states, so the fluctuations of η and of E are the largest there.

Short-range order. Figure 7 shows the Cowley parameters. If the only order were the long-range order, a Cu atom on α would find a Zn neighbour on β with probability $(1+\eta)/2$, and a Cu atom on β with probability $(1-\eta)/2$. Averaging over the two cases gives $\alpha_1 = -\eta^2$ and $\alpha_2 = +\eta^2$ (dashed lines). Below T_c the Cowley parameters follow these curves: the long-range order explains almost all the local order. Above T_c, however, $\langle\eta^2\rangle$ is almost zero while α_1 stays clearly negative: at 800 K, $\langle\alpha_1\rangle = -0.20$ whereas $-\langle\eta^2\rangle = -0.05$, and at 1600 K $\langle\alpha_1\rangle$ is still -0.07. Cu atoms keep more Zn first neighbours than in a random alloy: this *short-range order* survives the loss of long-range order and fades slowly with temperature, as observed experimentally in ordering alloys [12].

7 Transition temperature

We estimate T_c in three ways (Table 4): the maximum of the heat capacity, the maximum of the susceptibility, and the temperature where $\langle|\eta|\rangle$ drops the fastest (minimum of $\mathrm{d}\langle|\eta|\rangle/\mathrm{d}T$, by finite differences). The three methods give the same grid temperature, 720 K. The grid step is 10 K there, but the peaks are flat at the top, between 700 and 730 K (Table 5): we retain $T_c \approx 720$ K.

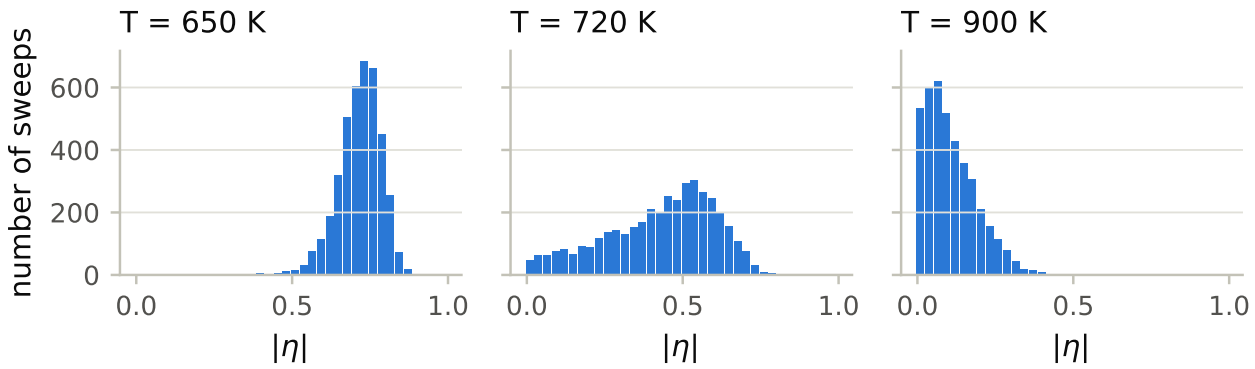


Figure 6: Histograms of the 4000 values of $|\eta|$ measured (one per sweep) below, near and above the transition. In our box $|\eta|$ can only take the values $2k/216$; each bin contains three of them (width $6/216 \approx 0.028$).

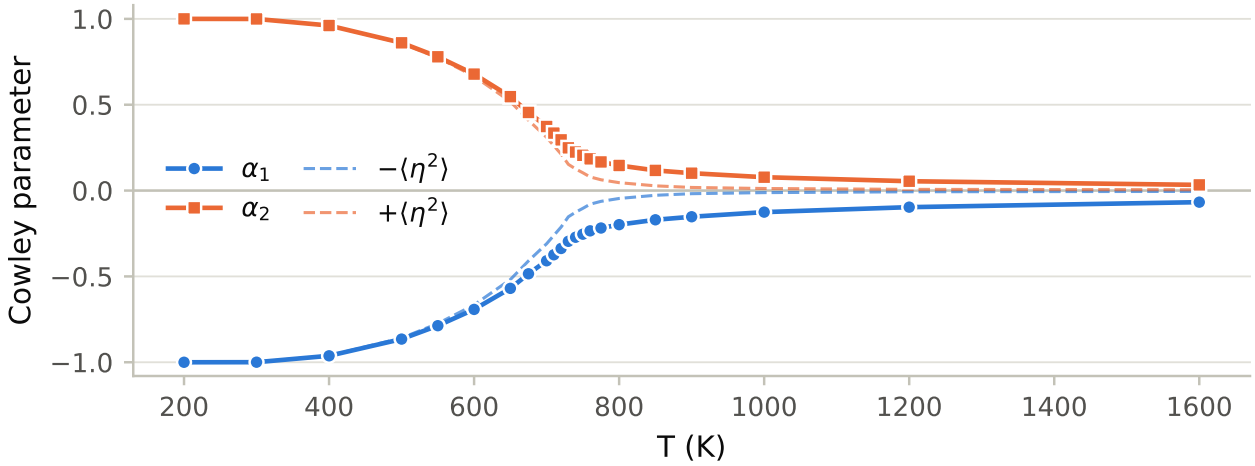


Figure 7: Cowley short-range order parameters for the first (α_1, circles) and second (α_2, squares) neighbour shells. The dashed lines $\mp\langle\eta^2\rangle$ are the values expected if the long-range order alone were present. The difference between symbols and dashed lines above 750 K is the short-range order.

Table 4: Transition temperature. The Monte Carlo values are read on the temperature grid of Table 2 (step 10 K between 700 and 760 K). The mean-field value is the Bragg–Williams approximation applied to the same pair model (see text).

Method	T_c (K)
Maximum of the heat capacity C	720
Maximum of the susceptibility χ	720
Steepest drop of $\langle\|\eta\|\rangle$	720
Bragg–Williams mean field, same model	894
Experiment (β-brass) [1–3]	$\approx$ 740–750

Comparison with mean field. The Bragg–Williams approximation [4] keeps the long-range order η but treats the atoms of each sublattice as independent. The energy then reads $E/N = E_{\mathrm{dis}}/N - \Delta E\,\eta^2$, with $\Delta E = (E_{\mathrm{dis}} - E_{\mathrm{B2}})/N = 0.0385\,\mathrm{eV/atom}$, and the entropy per atom is that of the random filling of each sublattice, $s \approx k_B(\ln 2 - \eta^2/2)$ for small η. The free energy $E/N - Ts$ develops a minimum at $\eta \neq 0$ below $k_B T_c = 2\Delta E$, that is $T_c^{\mathrm{MF}} = 894\,\mathrm{K}$. Our Monte Carlo value is about 20 % lower. The difference comes from the short-range order seen in Figure 7: above T_c the atoms can lower their energy by local arrangements without any long-range order, which mean field ignores. The disordered phase is therefore more stable than in mean field, and the transition happens at lower temperature.

8 Discussion

Agreement with experiment. Our estimates of T_c are close to the experimental value, which is quite good for a model with only six parameters and no adjustment. This agreement should not be over-interpreted: the model ignores the vibrations of the atoms, the thermal expansion of the lattice and the difference in size between Cu and Zn, and its pair energies are simple numbers. For comparison, a rigid-lattice Monte Carlo study of CuZn with energies computed from first principles finds $T_c \approx 870\,\mathrm{K}$, about 120 K above experiment [3]: a good T_c is not a proof that every ingredient of the model is right.

Finite size. With $L = 6$ the transition is rounded: $\langle|\eta|\rangle$ decreases continuously instead of vanishing at T_c, never goes below the floor of 0.04, and the peaks of C and χ have a finite height and width. The three estimators agree here, but they could differ by a few tens of kelvins: with only 800 sweeps per temperature, a first version of our analysis gave 700, 720 and 730 K. The position of the peaks also depends on L. A finite-size study with several box sizes (for example with the crossing of the Binder cumulant) would give T_c for an infinite crystal [9, 13].

Sampling near T_c. Successive measurements are correlated, especially near T_c where the order changes slowly (critical slowing down [14]). The heat capacity and the susceptibility are the noisiest quantities, because they are variances of correlated series; longer runs near T_c would make their peaks smoother.

Algorithm. Exchanging only first neighbours is a local dynamics that conserves the composition. It satisfies detailed balance, so it samples the correct equilibrium distribution, but it is slow to move order over long distances. Exchanges between any two sites would equilibrate faster at the same cost per step.

9 Conclusion

We wrote and tested a Metropolis Monte Carlo code for the order–disorder transition of CuZn on a rigid bcc lattice. The tests on the perfect B2 structure and on random lattices give the expected values, and the energy updated move by move stays exact. The cohesive energy of the B2 phase is 2.484 eV/atom. Starting from B2, the system needs at most about 12 000 steps to reach equilibrium, so we discarded 50 000 steps before measuring during 4000 sweeps. The energy, the heat capacity, the long-range order parameter and its susceptibility all show the transition at $T_c \approx 720\,\mathrm{K}$, close to the experimental value (about 740 K) and lower than the mean-field prediction (894 K). Above the transition, Cu and Zn still prefer to be first neighbours: this short-range order explains why mean field overestimates T_c. The main limits of this study are the small size of the box and the simple pair model; larger boxes and a finite-size analysis would be the next step.

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A Full results of the temperature scan

Table 5: Results of the temperature scan (conditions of Table 2). E: mean energy per atom (eV); $C/(Nk_B)$: heat capacity per atom; χ: susceptibility (eV^{-1}).

T (K)	$\langle E\rangle/N$	$C/(Nk_B)$	$\langle\lvert\eta\rvert\rangle$	χ	$\langle\alpha_1\rangle$	$\langle\alpha_2\rangle$
200	−2.4840	0.00	1.000	0	−1.000	1.000
300	−2.4840	0.01	1.000	0	−1.000	1.000
400	−2.4825	0.24	0.980	2	−0.962	0.961
500	−2.4788	0.59	0.926	6	−0.865	0.861
550	−2.4757	0.74	0.879	10	−0.787	0.779
600	−2.4720	0.94	0.815	17	−0.692	0.678
650	−2.4673	1.13	0.717	35	−0.570	0.547
675	−2.4639	1.40	0.631	86	−0.485	0.455
700	−2.4610	1.61	0.539	139	−0.409	0.373
710	−2.4596	1.44	0.487	166	−0.374	0.335
720	−2.4582	1.62	0.427	211	−0.337	0.295
730	−2.4565	1.18	0.350	205	−0.296	0.249
740	−2.4556	1.00	0.313	190	−0.272	0.224
750	−2.4549	0.86	0.276	183	−0.254	0.205
760	−2.4542	0.61	0.240	143	−0.235	0.185
775	−2.4535	0.45	0.212	116	−0.218	0.167
800	−2.4528	0.34	0.176	94	−0.199	0.146
850	−2.4517	0.20	0.133	58	−0.170	0.117
900	−2.4510	0.14	0.108	36	−0.152	0.101
1000	−2.4500	0.09	0.088	22	−0.126	0.078
1200	−2.4489	0.05	0.067	11	−0.097	0.054
1600	−2.4478	0.02	0.056	5	−0.068	0.033

B Code and reproduction

The model is written in Python in `src/model.py` (one module, about 190 lines):

- the array `sous_reseau` holds the species of each site (0 = Cu, 1 = Zn); site i of cell (x, y, z) and sublattice s has index $i = 2(x + 6y + 36z) + s$, so even sites are α and odd sites β;
- `FIRST_NEIGHBOURS` and `SECOND_NEIGHBOURS` are the neighbour tables, built once at import;
- `get_full_energy()` computes Equation (1); `energy_after_flip(E, i, j)` returns the energy after exchanging i and j from the local bonds only, without changing the configuration; `flipping_proba` and `flipping_action` apply the Metropolis rule;
- `cowley()` and `order_parameter()` return (α_1, α_2) and η;
- `metropolis_before_eq(n, T)` returns the energy after each of n steps from B2 (equilibration test); `metropolis(n_eq, n_meas, T, measure_every)` returns the series of E, $|\eta|$, α_1, α_2 measured after equilibration.

The analysis notebook `analysis.ipynb` uses these functions interactively. All the figures and tables of this report are produced by `report/make_figures.py`, with the parameters of Table 2. From the project folder:

```
python -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -B report/make_figures.py --rerun   # simulations, figures
cd report && latexmk -pdf report.tex slides.tex       # report and slides
```